

Aryan Saigal

AI/ML Engineer · Solutions Architect · Full-Stack Developer

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PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science student with hands-on production experience building RAG chatbots, multi-model OCR platforms, AI interview systems, and Text-to-SQL tools. Skilled in LangChain, FastAPI, Next.js, GCP Cloud Run, Docker, and MCP server integration. Passionate about building AI-powered products that solve real business problems.

EXPERIENCE

Solutions Architect Intern · [EMB Global \(Incuspaze\)](#)

May 2026 – Present

Gurgaon, India

- Designed and built AI/ML-powered tools across four project verticals — chatbots, document intelligence, interview automation, and SQL interfaces — using LangChain, FastAPI, Next.js and GCP Cloud Run.
- Integrated MCP (Model Context Protocol) servers for multi-modal AI generation and deployed production services via Docker, GitHub Actions CI/CD and GCP Cloud Run.
- Completed 8 Anthropic AI courses in parallel with project delivery, including Building with the Claude API, AI Fluency Framework and Advanced MCP (scored 9/10 in AI Fluency quiz).

ML Intern · [Innovate](#)

Aug 2025 – Oct 2025

Remote

- Developed a machine learning model for road traffic management titled 'Predictive Modeling for Road Traffic Management: A Data-Driven Approach', targeting congestion reduction through predictive analytics.
- Built end-to-end ML pipeline covering data collection, cleaning, feature engineering, model training (regression and classification), hyperparameter tuning and performance evaluation.
- Visualized model outputs and traffic patterns using Matplotlib and Seaborn, producing charts and reports that communicated insights to non-technical stakeholders.

Cloud Infrastructure Intern · [CloudSphere](#)

May 2025 – Jul 2025

Noida, India (Hybrid)

- Audited and documented cloud resources and dependencies across multi-cloud environments (AWS, GCP), mapping asset inventories and inter-service relationships.
- Analyzed compute, storage and network usage patterns to identify cost-saving opportunities; prepared optimization reports that highlighted underutilized resources.
- Assisted senior engineers in security assessments — identifying misconfigurations, IAM policy gaps and exposure risks — and configured performance monitoring alerts.
- Gained hands-on experience with cloud operations, infrastructure reporting and resource lifecycle management under senior engineer guidance.

Social Media & Content Intern · [The Cocoa Counter](#)

May 2024 – Jun 2024

New Delhi, India

- Managed brand social media accounts across platforms; developed monthly content calendars, wrote captions and collaborated with designers on visuals aligned to brand identity.
- Monitored platform engagement metrics, responded to customer inquiries and implemented feedback-driven changes that improved audience response rates.

PROJECTS

ConversAge AI – RAG Marketing & Content Suite [GitHub ↗](#)

May 2026

Stack: LangChain, FastAPI, Next.js, ChromaDB, MCP, OpenRouter, Vercel, Cloudflare

- Architected a full-stack RAG pipeline using LangChain and ChromaDB — documents (.pdf, .doc, .ppt) are chunked, embedded and stored as vectors; user queries retrieve the most relevant context before being passed to the LLM via OpenRouter.
- Extended the chatbot with five capabilities via MCP server integration: AI blog generation, image creation, video generation, a marketing campaign generator, and live web search — all accessible from a single Next.js chat interface.
- Secured the application with JWT-based authentication and session management; deployed frontend on Vercel and backend on Cloudflare Workers, with end-to-end testing across all file formats and generation pipelines.

Text-to-SQL Chatbot [GitHub ↗](#)

May 2026

Stack: FastAPI, LangChain, SQLite/PostgreSQL, Next.js

- Built a natural-language-to-SQL interface where users ask questions in plain English — the LLM generates the corresponding SQL query, which is validated and executed on a live database backend.
- Implemented a result-formatting layer that converts raw query output (rows, counts, aggregations) into clear, plain-language summaries — making data accessible to non-technical users without any SQL knowledge.

- Designed the system to handle multi-step queries, ambiguous phrasing, and schema-awareness by injecting table structure into the prompt context, reducing hallucinated or invalid SQL.

OCR Multi-Model Parser Platform [GitHub ↗](#)

May – Jun 2026

Stack: FastAPI, LlamaParse, Docling, LiteParser, GCP Cloud Run, Docker, GitHub Actions

- Built a document intelligence platform supporting four OCR engines (LlamaParse, Docling, LiteParser, Document AI) selectable per request — enabling comparison of extraction quality across different document types and layouts.
- Added an extract-page route for targeted page-level parsing, an admin dashboard for model configuration and usage monitoring, and full settings and documentation pages covering every parser's parameters.
- Containerized with Docker, deployed to GCP Cloud Run for auto-scaling, and set up GitHub Actions CI/CD for automated build and release; additionally published a standalone CLI tool allowing users to run OCR locally without the web interface.

AI Web Interview Platform [GitHub ↗](#)

Jun 2026

Stack: Next.js, FastAPI, OpenAI Realtime Voice API, WebRTC

- Built a browser-based AI interviewer using OpenAI's Realtime Voice API — the system conducts spoken interviews, listens to candidate responses in real time, and generates follow-up questions dynamically based on answers.
- Integrated WebRTC for screen sharing and implemented fullscreen enforcement and candidate movement/activity recording via webcam analysis, providing proctoring signals to detect suspicious behaviour during sessions.
- Developed an admin panel for creating and configuring interview sessions — including question sets, time limits, and candidate access controls — with session recordings and activity logs stored for post-interview review.

KANOON – AI-Powered Legal Decision Support System [GitHub ↗](#)

May – Jul 2025

Stack: Python, FastAPI, ChromaDB, React 18, spaCy, NLTK, Multilingual NLP (11 Indian languages)

- Designed a RAG pipeline over a curated Indian legal corpus covering IPC sections, FIR procedures and cyber laws — chunks are embedded and retrieved from ChromaDB to ground LLM responses in actual legal text, reducing hallucination.
- Implemented multilingual NLP support across 11 Indian languages using spaCy and NLTK for tokenization, transliteration and language detection, making legal information accessible to non-English-speaking users.
- Delivered a full-stack system (React 18 frontend, FastAPI backend) with a CLI tool for dataset ingestion, and authored an accompanying research paper on the application of AI/ML in Indian legal technology.

Local AI Coding Assistant (Ollama + Qwen Coder) [GitHub ↗](#)

Jun 2026

Stack: Ollama, Qwen2.5-Coder, Node.js, Supabase, JavaScript

- Built a local AI coding chatbot running entirely on-device using Ollama and the Qwen2.5-Coder model — no external API calls, full privacy, zero inference cost.
- Developed a Node.js backend with streaming responses and a browser-based UI; integrated Supabase authentication and persistent chat history across sessions.
- Demonstrated practical MLOps and local LLM deployment skills by packaging the model setup, backend and frontend into a single runnable project.

Customer Churn MLOps Pipeline [GitHub ↗](#)

2026

Stack: Python, XGBoost, MLflow, FastAPI, Docker, GitHub Actions, AWS

- Built an end-to-end MLOps pipeline for customer churn prediction using XGBoost — covers data preprocessing, feature engineering, model training, hyperparameter tuning and evaluation.
- Integrated MLflow for experiment tracking and model registry; exposed predictions via a FastAPI inference endpoint containerized with Docker.
- Automated CI/CD with GitHub Actions for model retraining and deployment, with monitoring hooks and AWS infrastructure for scalable production serving.

E-Commerce Sales Analysis Dashboard [GitHub ↗](#)

Aug – Oct 2024

Stack: Python (Pandas, NumPy, Matplotlib), Power BI, Excel

- Processed and cleaned a large e-commerce dataset using Pandas — handled missing values, normalized categories, engineered time-based features (month, season, day-of-week) and aggregated metrics by product, city and payment method.
- Identified key business insights: top-revenue product categories, highest-performing cities, peak seasonal sales windows and preferred payment methods — findings presented as an executive summary with supporting visualizations.
- Built an interactive Power BI dashboard and an Excel report with pivot tables and charts, enabling stakeholders to slice data by region, category and time period without requiring technical knowledge.

SKILLS

AI / ML	LangChain, RAG Pipelines, ChromaDB, OpenRouter, MCP Servers, LlamaParse, Docling, spaCy, NLTK, Generative AI
Backend	FastAPI, Python, Node.js, REST APIs, SQL, PostgreSQL, SQLite, Supabase
Frontend	Next.js 14, React 18, HTML5, CSS3, JavaScript, Tailwind CSS
Cloud & DevOps	GCP Cloud Run, Docker, GitHub Actions CI/CD, Vercel, Cloudflare
Data & Analytics	Pandas, NumPy, Matplotlib, Power BI, Excel
Other	Cyber Security, AWS Cloud Foundations, Networking (TCP/IP, IPv4, Routing)

EDUCATION

B.Tech CSE · Amity University, Noida	2022 – 2026 · GPA 7.2
Class XII CBSE	2022 · 91%
Class X CBSE	2020 · 85%

COURSES & CERTIFICATIONS

Anthropic AI Courses:

- Building with the Claude API – May 2026
- AI Fluency: Framework & Foundations (Quiz: 9/10) – May 2026
- AI Capabilities and Limitations – May 2026
- Claude 101 – Jun 2026
- Introduction to Claude Cowork – Jun 2026
- Introduction to Model Context Protocol – Jun 2026
- Introduction to Agent Skills – Jun 2026
- Model Context Protocol: Advanced Topics – Jun 2026

Other Certifications:

- AWS Academy Cloud Foundations
- Python Essentials 1 & 2 (Cisco)
- Introduction to Cybersecurity (Cisco)
- Introduction to Modern AI / Data Science / Generative AI
- Introduction to Probability and Statistics

LANGUAGES

English (Full Professional) · Hindi (Native) · French (Read / Write)